Zhe (Jay) Xu

Research Scientist @ Meta
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Education

University of Illinois Urbana-Champaign — Champaign, IL, USA
Ph.D. in Computer Science Thesis: Optimal graph learning — *May 2021 – April. 2025*

University of Illinois Urbana-Champaign — Champaign, IL, USA
M.S. in Computer Science Thesis: Dense subgraph detection on multi-layered networks — *Aug. 2019 – May 2021*

Fudan University — Shanghai, China
B.E. in Electronic Engineering — *Sep. 2014 – Jun. 2018*

Publication

Preprint Papers

P1. Xinyu Zhu, **Zhe Xu**, Xiaohan Wei, Yunchen Pu, Fei Tian, Chonglin Sun, Kaushik Rangadurai, et al. Self-Guided Test-Time Training for Long-Context LLMs. Under review.

P2. **Zhe Xu**, Ankit Peshin, Chiyu Zhang, Feng Qi, Johnson Lui, Anil Ramakrishna, et al. Tokenizing Numerical and Embedding Features for LLM RecSys. Under review.

P3. Yufei Li, Zaiwei Zhang, Mingfu Liang, Kavosh Asadi, **Jay Xu**, Jimmy Kim, Chongyang Bai, Jieyi Zhang, et al. GR2 Technical Report. Under review.

P4. Mingfu Liang*, Yufei Li*, **Jay Xu***, Kavosh Asadi, Xi Liu, et al. Generative Reasoning Re-ranker. Under review.

P5. Beidi Zhao, Boxin Du, **Zhe Xu**, Liangyue Li, Hanghang Tong. Learning Optimal Propagation for Graph Neural Networks. Under review.

P6. Yuchen Yan, Yuzhong Chen, Huiyuan Chen, Xiaoting Li, **Zhe Xu**, Mahashweta Das, Hanghang Tong. THiGCN: Temporal Heterophilic Graph Convolutional Network for Event-based Continuous Graphs. Under review.

Accepted Conference Papers

C1. **Zhe Xu**, Kaveh Hassani, Si Zhang, Hanqing Zeng, Michihiro Yasunaga, Limei Wang, Dongqi Fu, Ning Yao, Bo Long, Hanghang Tong. Language Models are Graph Learners. Findings of EACL 2026.

C2. **Zhe Xu**, Menghai Pan, Yuzhong Chen, Huiyuan Chen, Yuchen Yan, Mahashweta Das, Hanghang Tong. Fine-grained Graph Rationalization. CIKM 2025.

C3. **Zhe Xu**, Ruizhong Qiu, Yuzhong Chen, Huiyuan Chen, Xiran Fan, Menghai Pan, Zhichen Zeng, Mahashweta Das, Hanghang Tong. Discrete-state Continuous-time Diffusion for Graph Generation. NeurIPS 2024.

C4. **Zhe Xu**, Yuzhong Chen, Menghai Pan, Huiyuan Chen, Mahashweta Das, Hao Yang, Hanghang Tong. Kernel Ridge Regression-Based Graph Dataset Distillation. SIGKDD 2023.

C5. **Zhe Xu**, Yuzhong Chen, Qinghai Zhou, Yuhang Wu, Menghai Pan, Hao Yang, Hanghang Tong. Node Classification Beyond Homophily: Towards a General Solution. SIGKDD 2023.

C6. **Zhe Xu**, Kaize Ding, Yu-Xiong Wang, Huan Liu, Hanghang Tong. Generalized Few-Shot Node Classification on Graphs. ICDM 2022. **(Best-Ranked Papers)**.

C7. **Zhe Xu**, Boxin Du, Hanghang Tong. Graph Sanitation with Application to Node Classification. TheWebConf 2022.

C8. **Zhe Xu**, Si Zhang, Yinglong Xia, Liang Xiong, Jiejun Xu, Hanghang Tong. DESTINE: Dense Subgraph Detection on Multi-Layered Networks. CIKM 2021.

C9. **Zhe Xu**, Si Zhang, Yinglong Xia, Liang Xiong, Hanghang Tong. Ranking on Network of Heterogeneous Information Networks. IEEE BigData 2020.

C10. Dongqi Fu*, **Zhe Xu***, Bo Li, Hanghang Tong, and Jingrui He. A View-Adversarial Framework for Multi-View Network Embedding. CIKM 2020. (*equal contribution)

C11. Ruizhong Qiu, **Zhe Xu**, Wenxuan Bao, Hanghang Tong. Turing Completeness of Prompting. ICLR 2025.

C12. Huiyuan Chen, **Zhe Xu**, Chin-Chia Michael Yeh, Vivian Lai, Yan Zheng, Minghua Xu, Hanghang Tong. Masked Graph Transformer for Large-Scale Recommendation. SIGIR 2024.

C13. Qinghai Zhou, Yuzhong Chen, **Zhe Xu**, Yuhang Wu, Menghai Pan, Mahashweta Das, Hao Yang, Hanghang Tong. Graph Anomaly Detection with Adaptive Node Mixup. CIKM 2024.

C14. Zhichen Zeng, Ruizhong Qiu, **Zhe Xu**, Zhining Liu, Yuchen Yan, Tianxin Wei, Lei Ying, Jingrui He, Hanghang Tong. Graph Mixup on Approximate Gromov–Wasserstein Geodesics. ICML 2024.

C15. Haiyun Jiang, Li Cui, **Zhe Xu**, Deqing Yang, et al. Relation Extraction Using Supervision from Topic Knowledge of Relation Labels. IJCAI 2019.

C16. Dongqi Fu*, Liri Fang*, Zihao Li*, **Zhe Xu***, Hanghang Tong, Vetle I Torvik, Jingrui He. Bring Complex Geometric Information to LLMs: A Positional Survey of Graph Parametric Representation. LOG 2025. (*equal contribution)

C17. Haobo Xu, Yuchen Yan, Dingsu Wang, **Zhe Xu**, Zhichen Zeng, Tarek F. Abdelzaher, Jiawei Han, Hanghang Tong. SLOG: An Inductive Spectral Graph Neural Network Beyond Polynomial Filter. ICML 2024.

C18. Xinyu He, Chenhan Xiao, Haoran Li, Ruizhong Qiu, **Zhe Xu**, Yang Weng, Jingrui He, Hanghang Tong. PowerGrow: Feasible Co-Growth of Structures and Dynamics for Power Grid Synthesis. SIGKDD 2026.

C19. Zhining Liu, Ruizhong Qiu, Zhichen Zeng, Hyunsik Yoo, David Zhou, **Zhe Xu**, Yada Zhu, Kommy Weldemariam, Jingrui He, Hanghang Tong. Class-Imbalanced Graph Learning without Class Rebalancing. ICML 2024.

Accepted Journal Papers

J1. **Zhe Xu**, Kaize Ding, Yu-Xiong Wang, Huan Liu, Hanghang Tong. Generalized Few-Shot Node Classification on Graphs: Toward an Uncertainty-Based Solution. KAIS 2023.

J2. Kaize Ding, **Zhe Xu**, Hanghang Tong, Huan Liu. Data Augmentation for Deep Graph Learning: A Survey. SIGKDD Explorations, 2022.

J3. Qing Chen, Nan Chen, Guande Wu, Ziyan Liu, **Zhe Xu**, Hanghang Tong, Nan Cao. Calliope-Net: Automatic Generation of Graph Data Facts via Annotated Node-link Diagrams. IEEE Transactions on Visualization and Computer Graphics 2023.

Talks

T1. *Lecture-Style Tutorial*: Natural and Artificial Dynamics in GNNs: A Tutorial. WSDM 2023.

T2. *Invited Talk*: Graph Data Distillation. Visa Research. July 2023.

Research Experience

Research Scientist — Sunnyvale, CA, USA
Monetization, Meta. Manager: Dr. Wen-Yen Chen. — *Mar. 2025 – Present*
- Contribute to the ads recommender system at Meta.

Research Intern — Sunnyvale, CA, USA
Monetization, Meta. Manager: Dr. Si Zhang. — *May. 2024 – Nov. 2024*
- Research on the interaction between large language models and graph machine learning.

PhD Intern — Palo Alto, CA, USA
Visa Research, Visa Inc. Manager: Dr. Menghai Pan. — *May. 2023 – Aug. 2023*
- Conducted research on graph data augmentation in the life cycle of graph machine learning.

Research Intern — New York City, NY, USA
Applied Machine Learning Team, Meta. Manager: Kelvin Niu. — *May. 2022 – Aug. 2022*
- Constructed a large-scale representation learning system on user-video interaction data via graph neural networks.

Algorithm Engineer Intern Hangzhou, Zhejiang, China
Search Department, Alibaba Group. Manager: Dr. Muhua Zhu. *Jun. 2018 – Aug. 2018*

- Constructed E-commerce knowledge graphs on specific shopping topics.

Graduate Research Assistant Urbana, IL, USA
University of Illinois Urbana-Champaign. Advisor: Prof. Hanghang Tong. *Aug. 2019 – Present*

- Conducted research on augmenting graph data

Graduate Research Assistant Tempe, AZ, USA
Arizona State University. Advisor: Prof. Hanghang Tong. *Aug. 2018 – Aug. 2019*

- Conducted research on multi-layered networks.

Undergraduate Research Assistant Shanghai, China
Fudan University. Advisor: Prof. Yanghua Xiao. *Sep. 2016 – Jun. 2018*

- Constructed a financial product knowledge base and a financial product Q&A system (Chinese).

Grant Experience

- Helped a successful NSF proposal on 'SLES: NetSafe: Towards a Computational Foundation of Safe Graph Neural Networks.'
- Helped a successful NSF proposal on 'Collaborative Research: Towards a Theoretic Foundation for Optimal Deep Graph Learning.'

Teaching Experience

Teaching Assistant Champaign, IL, USA
CS 514 Advanced Topics in Network Science *Fall 2023*

Teaching Assistant Champaign, IL, USA
CS 512 Data Mining Principle *Fall 2022, Spring 2021*

Teaching Assistant Champaign, IL, USA
CS 412 Introduction to Data Mining *Fall 2021*

Service

Conference reviewer: AISTATS (2025), ICML (2024), SDM (2024), ICLR (2024-2025), AAAI (2023-2025), NeurIPS (2023-2024), KDD (2023,2025), CIKM (2021-2025), TheWebConf (2023-2025).
Journal reviewer: TNNLS, TVCG, TKDD.
Workshop reviewer: GLFrontiers (2024), TrustLOG (2022)
Conference subreviewer: ICDCS (2022), SIGIR (2021), CIKM (2019), ICDCS (2019).